

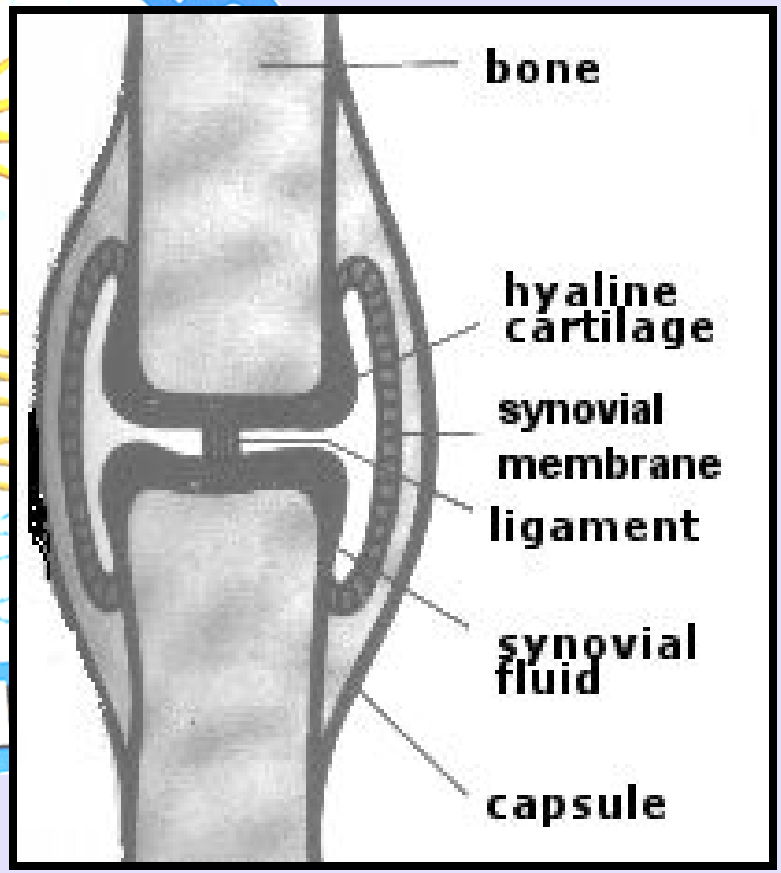
GENERAL CONCEPTS OF JOINTS

LEARNING OBJECTIVE

- **At the end of lecture student should be able to:**
  - Define the joints
  - Classify the joints on the basis of uniting material
  - Define a synovial joint
  - Enlist the features of synovial joint
  - Classify Synovial joints on the basis of:
    - Shape of articulating surfaces
    - Degree of mobility
  - Express the principles of innervation and blood supply of synovial joints

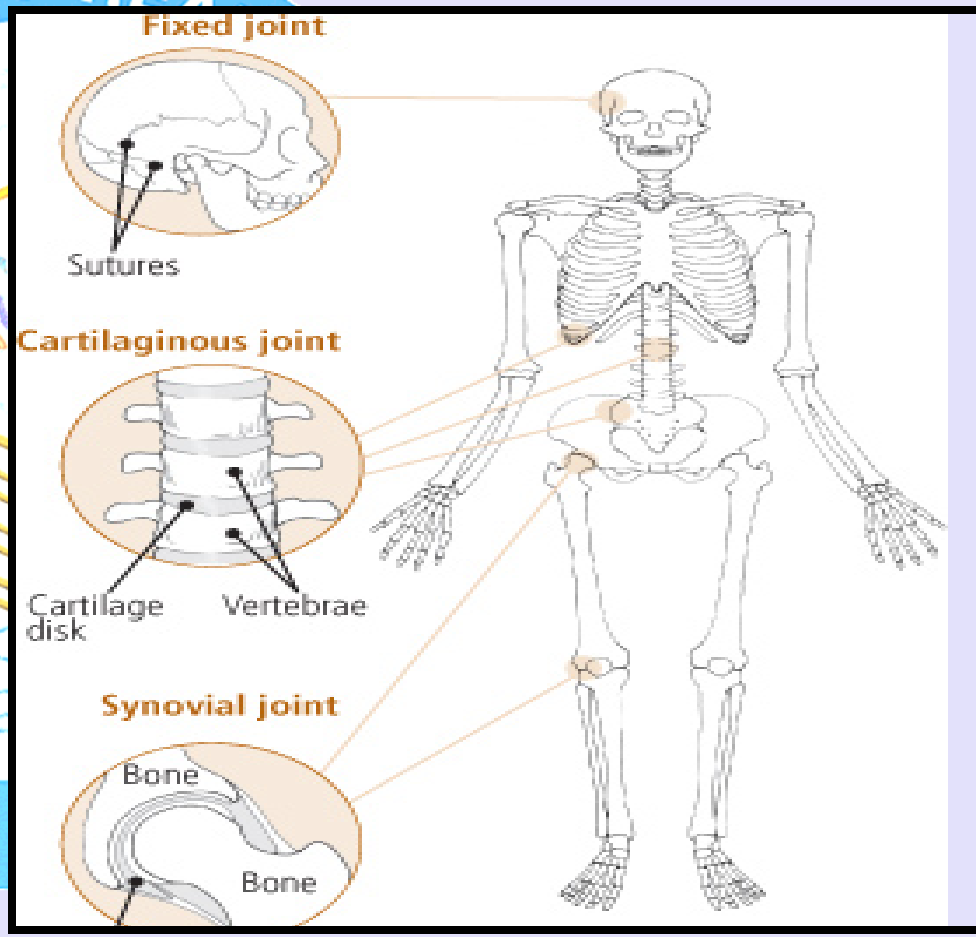
JOINT

- ***A joint is the location at which two or more <u>bones</u> make contact.***
- ***They are constructed to allow movement and provide mechanical support, and are classified structurally and functionally***



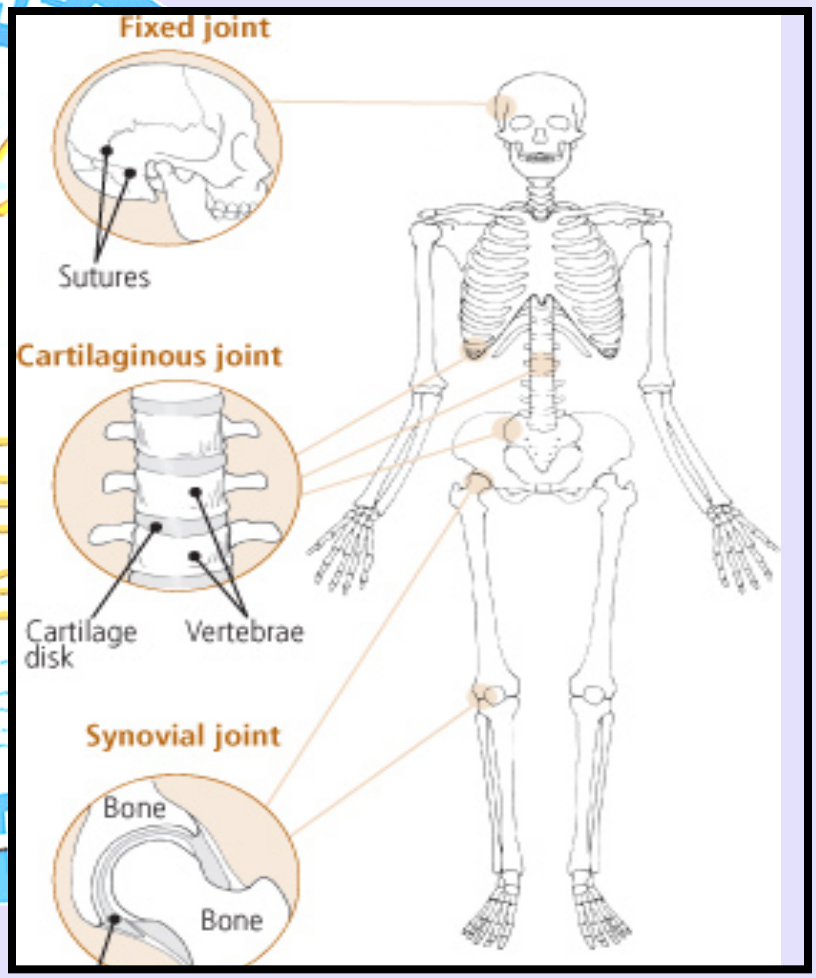
CLASSIFICATION OF JOINTS

- *Joints are classified*
  - *Structurally*
  - *functionally.*
- *Structural classification is determined by how the bones connect to each other.*
- *Functional classification is determined by the degree of movement between the articulating bones*



STRUCTURAL CLASSIFICATION

- ***There are three structural classifications of joints:***
- **Fibrous joint -** joined by fibrous connective tissue
- ***Cartilaginous joint -*** joined by cartilage
- ***Synovial joint -*** not directly joined



FUNCTIONAL CLASSIFICATION

➢ ***Joints can also be classified functionally***

- **Synarthrosis -** permits little or no mobility. Most joints are fibrous joints (e.g., skull sutures, syndesmosis: inferior tibiofibular joint).
- **Amphiarthrosis -** permits slight mobility. Most joints are cartilaginous joints(e.g., vertebrae).
- **Diarthrosis -** permits a variety of movements. All diarthrosis joints are synovial joints (e.g., shoulder, hip, elbow, knee, etc.), and the terms "diarthrosis“.

BIOMECHANICAL CLASSIFICATION

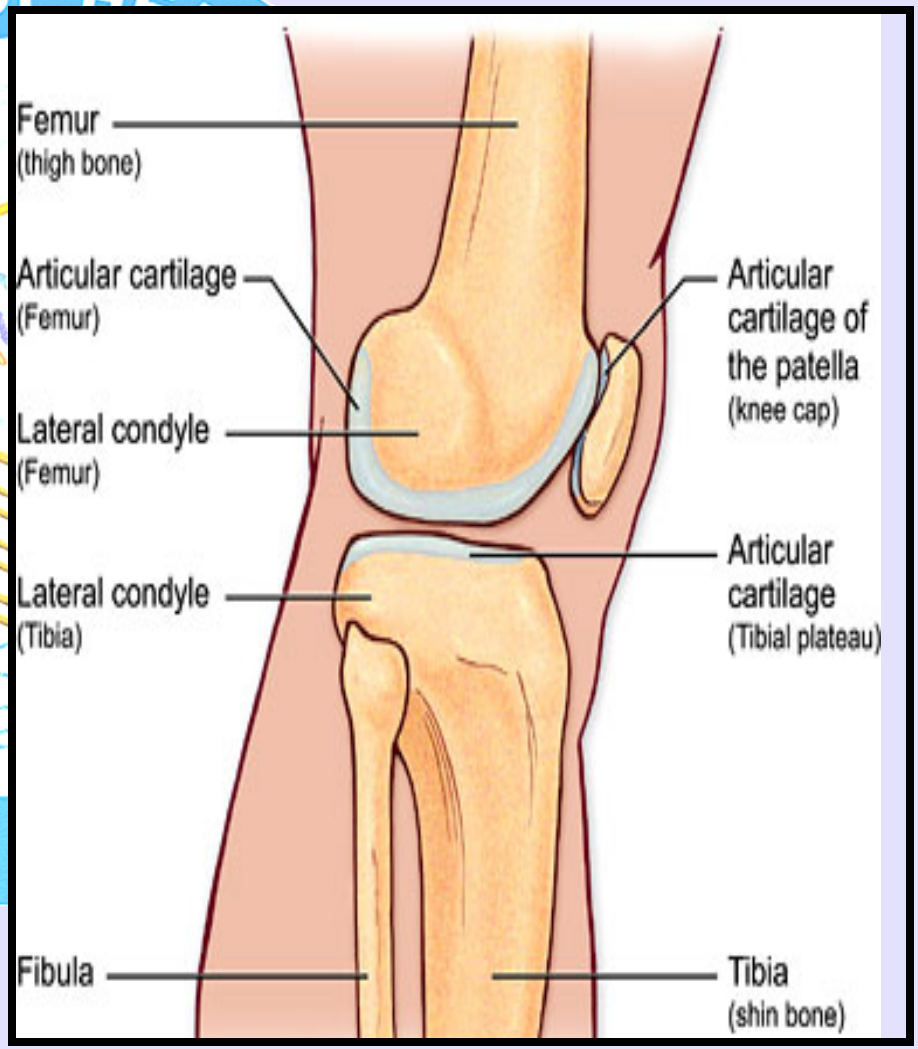
➢ ***Joints can also be classified based on their anatomy or on their biomechanical properties.***

- **According to the anatomic classification**
  - Simple joints
  - Compound joints
  - complex joints
    - ***Simple Joint:***
      - 2 articulation surfaces (eg. shoulder joint, hip joint)
    - ***Compound Joint:***
      - 3 or more articulation surfaces (eg. radiocarpal joint)
    - ***Complex Joint:***
      - 2 or more articulation surfaces and an articular disc or meniscus (eg. knee joint)

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SYNOVIAL JOINT

- *Is the most common and most movable type of joint .*
- *As with most other joints, synovial joints achieve movement at the point of contact of the articulating bones.*

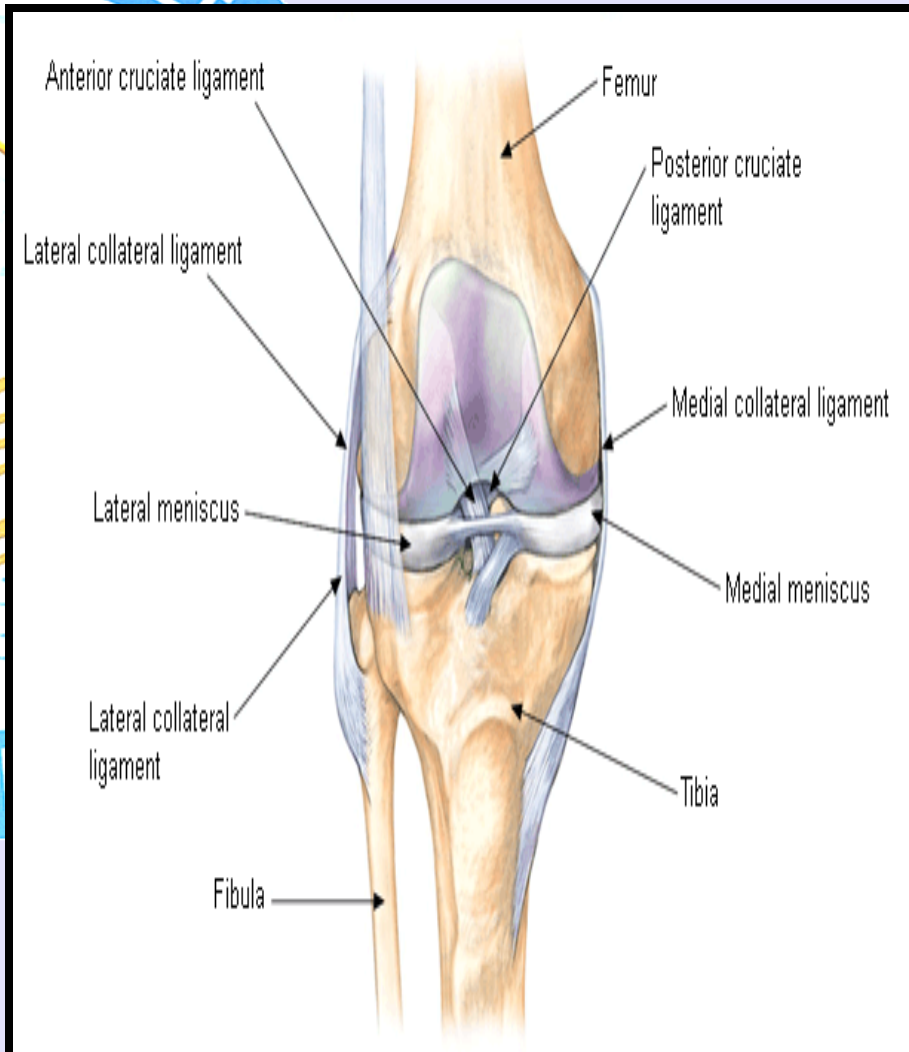


STRUCTURAL AND FUNCTIONAL DIFFERENCES

- *The main structural differences between synovial and fibrous joints is the existence of capsules surrounding the articulating surfaces of a synovial joint and the presence of lubricating synovial fluid within that capsule (synovial cavity).*

FEATURES OF SYNVIAL JOINTS

- **Articular capsule***:* The fibrous capsule is continuous with the periosteum of bone. It is also highly innervated but avascular (lacking blood and lymph vessels)
- **Articular cartilage***:* lines the epiphyses of joint end of bone. Provides the loading and unloading mechanism to resist load and shock
- **Synovial membrane***:* the inner layer of the fibrous articular capsule. The synovial membrane covers the lining of the synovial cavity where articular cartilage is absent.



CLASSIFICATION OF SYNOVIAL JOINTS

- **Synovial joints are classified on th basis of:**
  - Shape of articulating surface
  - Degree of mobility

- Gliding joint
- Hinge joint
- Pivot joint
- Condyloid joints
- Saddle joints
- Ball and socket joints
- Ellipsoid joints.

GLIDING JOINT (PLANE JOINTS)

- ***These joints allow only gliding or sliding movements.***

Examples:

- Carpals of the wrist
- acromioclavicular joint.

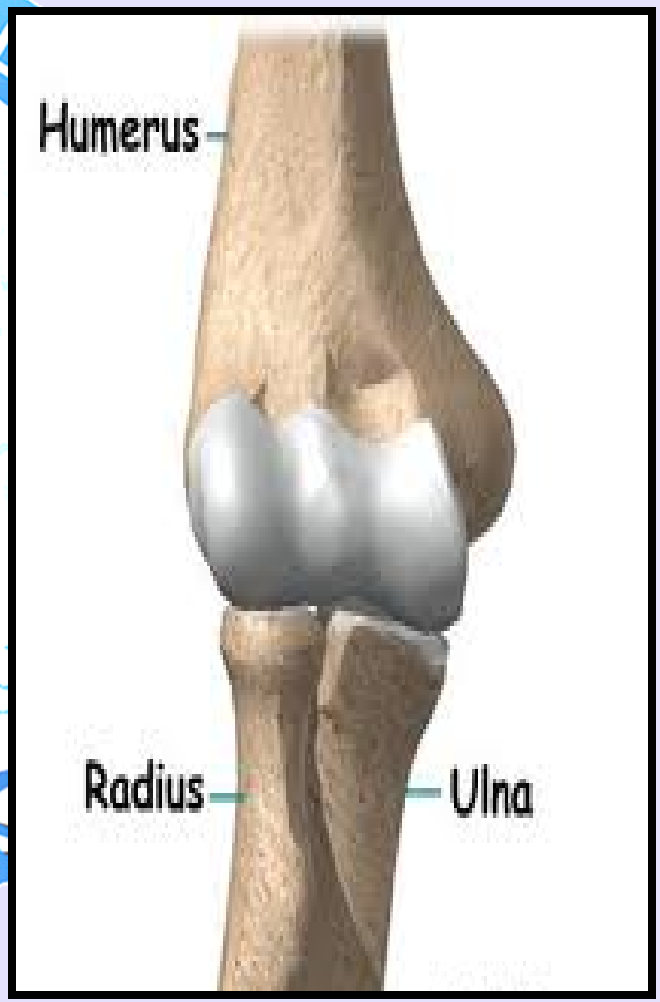


HINGE JOINT

- ***These joints act like a door hinge, allowing flexion and extension in just one plane.***

Example:

- The elbow (between the humerus and the ulna).

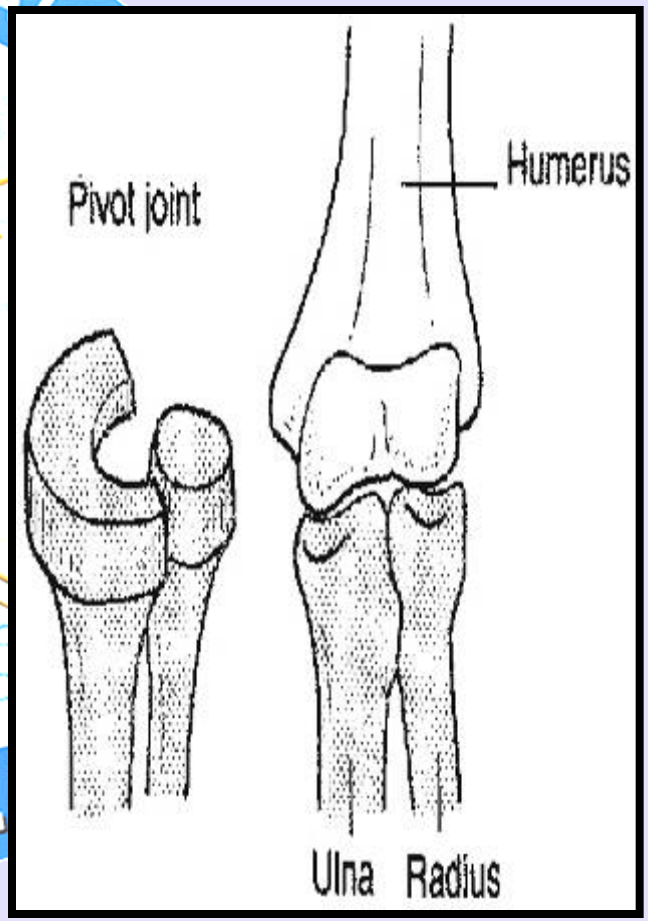


PIVOT JOINTS

- ***There is central bony pivot surrounding by osteocartilaginous ring one bone rotates about another***

Example:

- Atlanto-axial joint
- proximal radioulnr joint
- distal radioulnar joint.

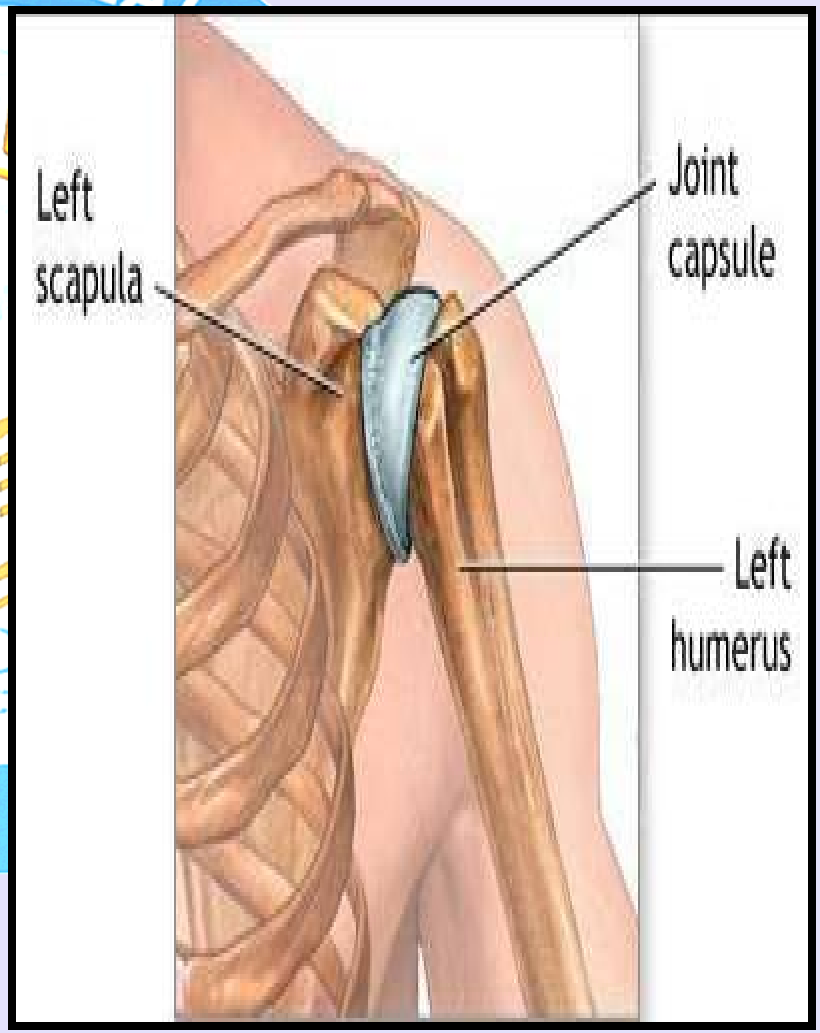


CONDYLOID JOINT

- *(A condyloid joint is where two bones fit together with an odd shape (e.g. anellipse).*
- *one bone is concave, the other convex*

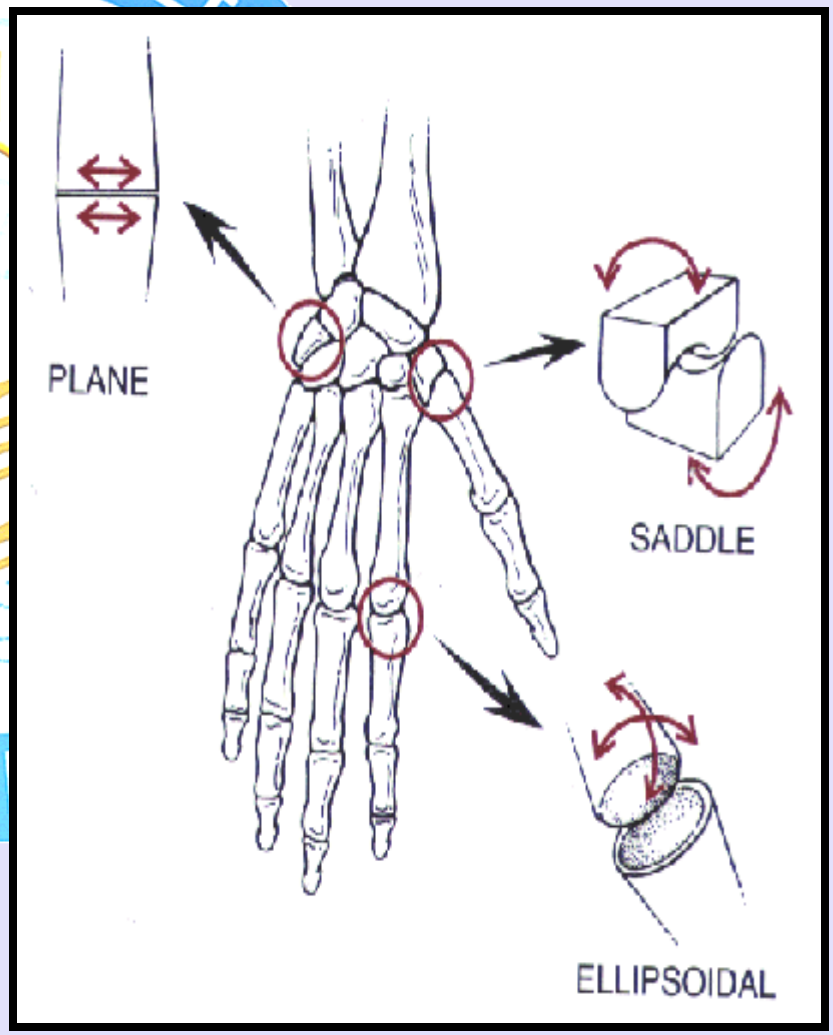
Example:

- *Knee joint*
- *Temporomandibular joint*



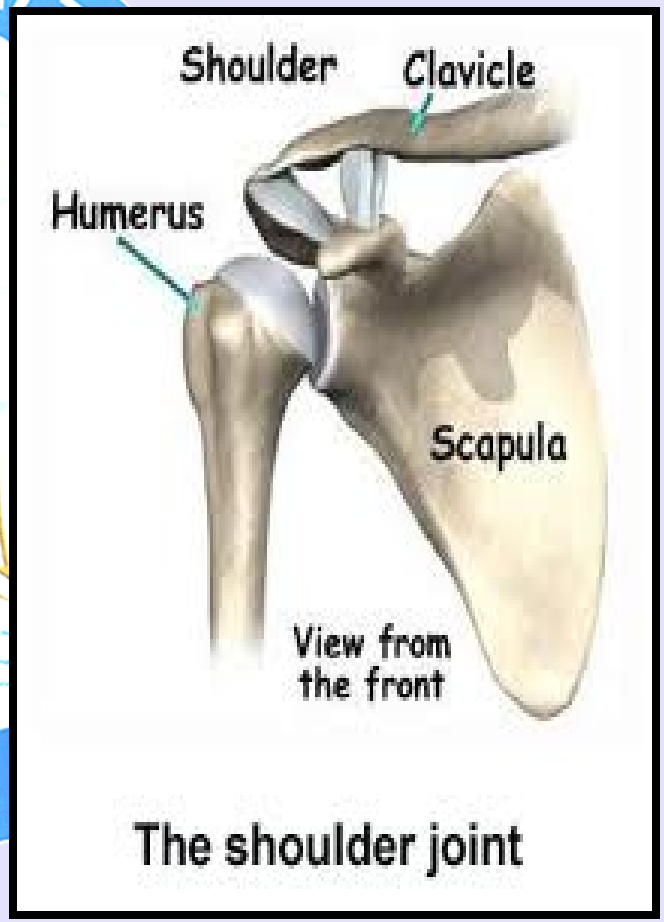
SADDLE JOINT

- ***Saddle joints, which resemble a saddle,in this articular surface of the bones are reciprocally concavoconvex***
- ***Examples:***
  - ***1st Carpometacarpal i.e. Joint of thumb***
  - ***sternoclavicular joint.***
  - ***Calcaniocuboid joint***



BALL AND SOCKET JOINTS

- ***These allow a wide range of movement***
- ***Example:***
  - ***The shoulder i.e. (glenohumeral) and***
  - ***hip joints.***



ELLIPSOID JOINTS

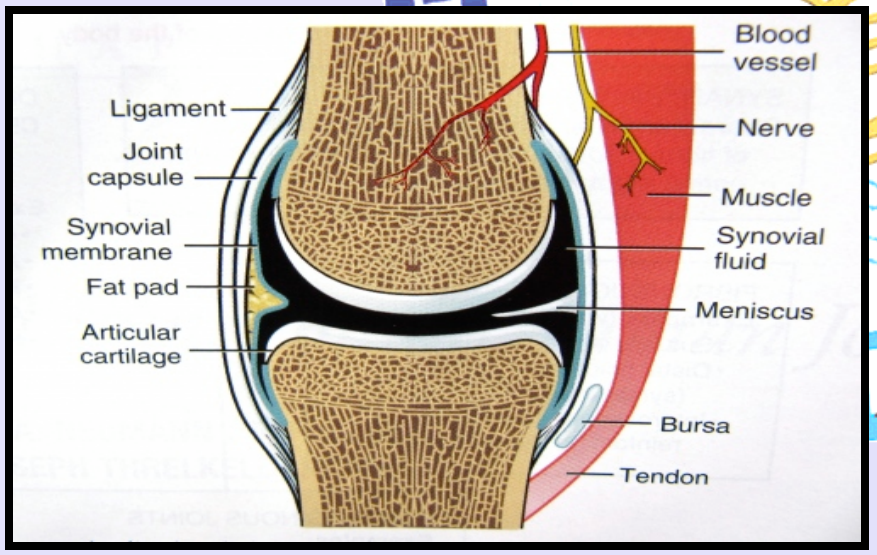
Example:

Wrist joint



NERVE SUPPLY OF SYNOVIAL JOINT

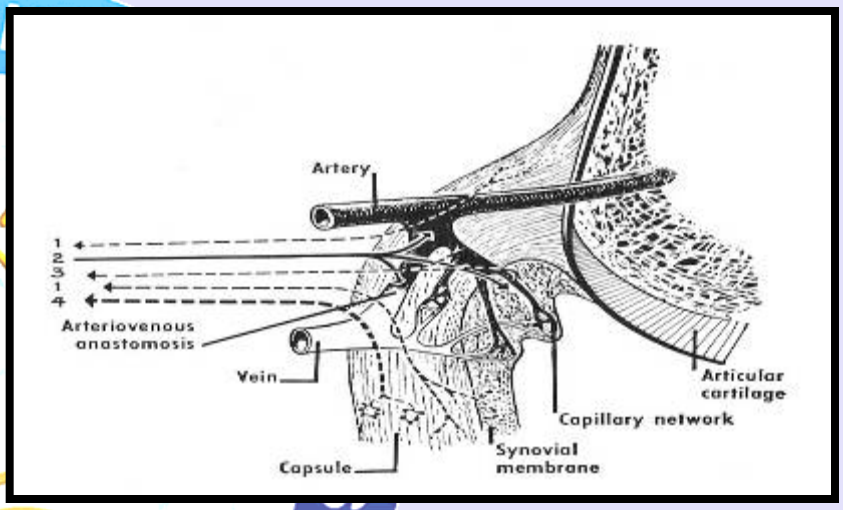
- ***It is derived from the nerve supply of muscles acting on the joint.***

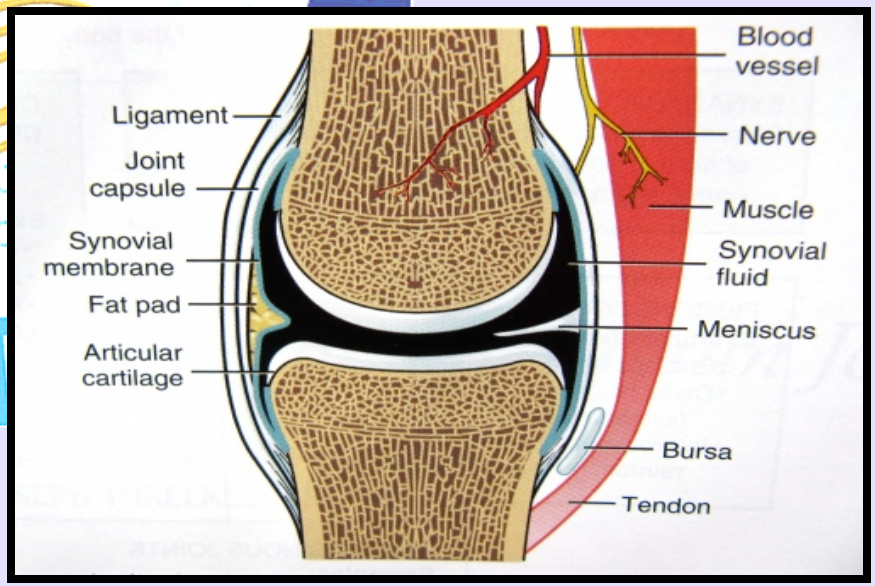




BLOOD SUPPLY OF SYNOVIAL JOINT

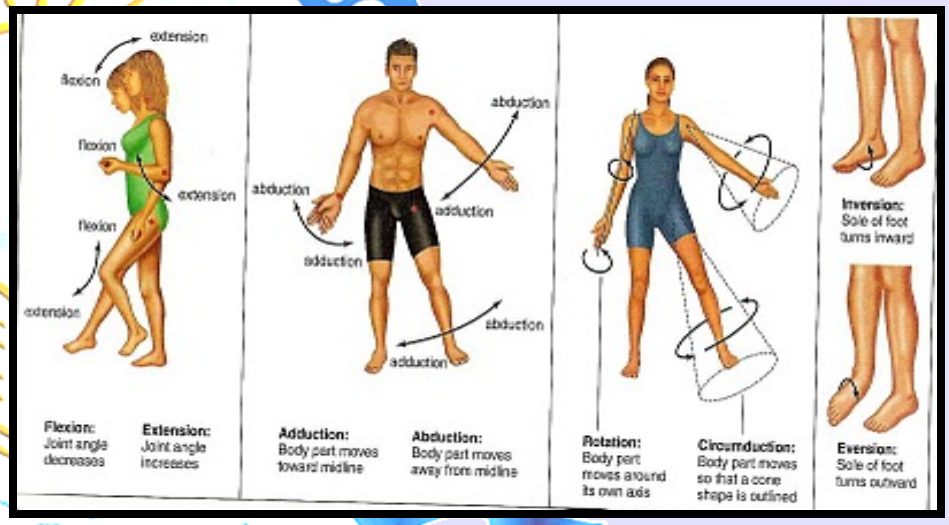
- ***From the arteries sharing in the anastomosis around the joint (neighboring vessels).***





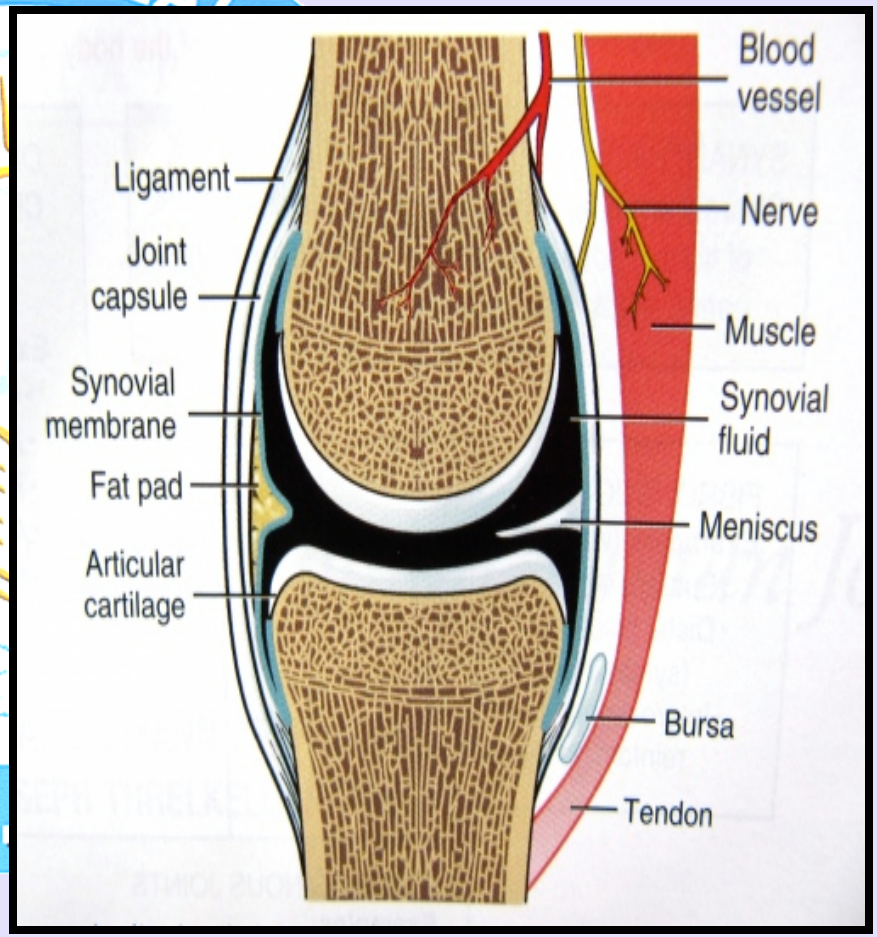
MOVEMENTS POSSIBLE

- *The movements possible with synovial joints are:*
- *Abduction: movement away from the mid-line of the body.*
- *Adduction: movement towards the mid-line of the body.*
- *Extension: straightening limbs at a joint.*
- *Flexion: bending the limbs at a joint.*
- *Rotation: a circular movement around a fixed point.*



FACTORS INFLUENCING JOINT STABILITY

- *The shape of articular surfaces.*
- *capsule & ligaments.*
- *Muscle Tone.*
- *gravity*
- *Atmospheric Pressure.*



REFERENCES

- ***CLINICALLY ORIENTED ANATOMY***

  Keith L. Moore

  6th edition

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THANK YOU